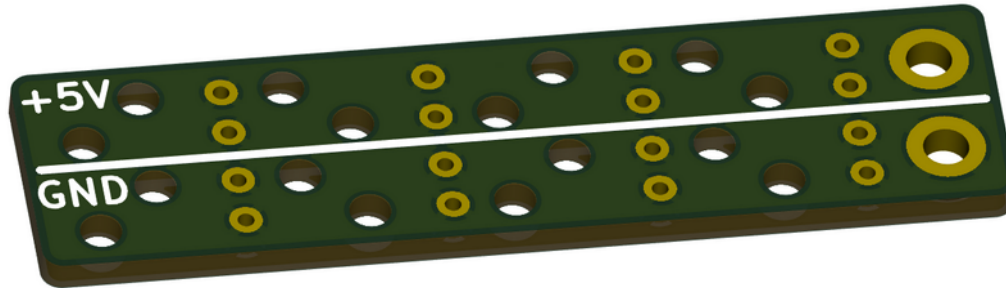
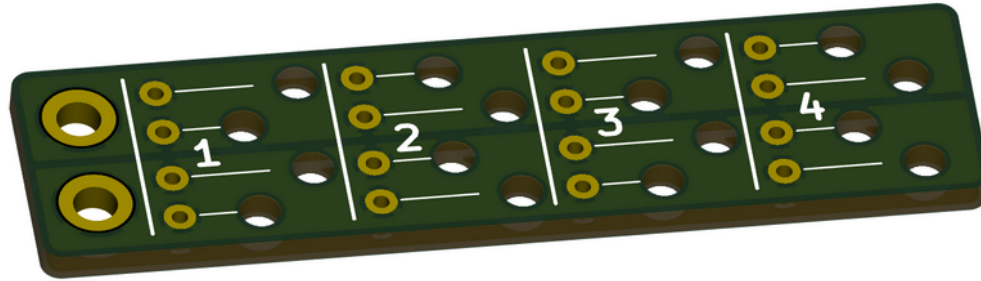


POWER FANOUT (2 POLE)



☕ 12 mm X 47 mm

☕ Fans out black (Bk) and white (W) wires from two 12-AWG lines to sixteen 22-AWG jumpers (12" long with other end precrimped) w/ 1cm spacing

☕ Includes strain relief holes for the more fragile 22-AWG wires

☕ Makes batch soldering easier --- thread all 16 at once, strip and install all 16, then solder all 16

☕ Heatshrink whole board after assembly

☕ Total boards needed: 36

☕ EFWD (4*4), EBWD (4*3), BKLM (4*2)

2, 3, & 4-WAY 12 AWG FANOUT

☕ WAGO butt splices are super easy to use
and can be disconnected later if problems
arise

☕ Need 24 total

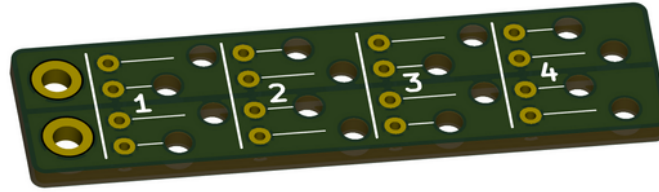


BASIC IDEA (FOR 1 OF 36)

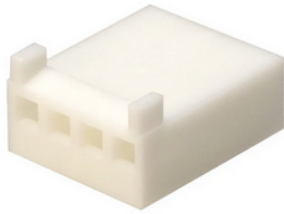
Strip and clamp
1w/1bk 12 AWG



Solder (2x)



Thread, strip 2-3
mm, insert, and
solder (16x)



Snap 2w/2bk crimped
terminals into connector (4x)



Stop ends from fraying
with two pieces of heat
shrink (8x)



Gather
2w/2bk &
cover with
sheath
(4x)



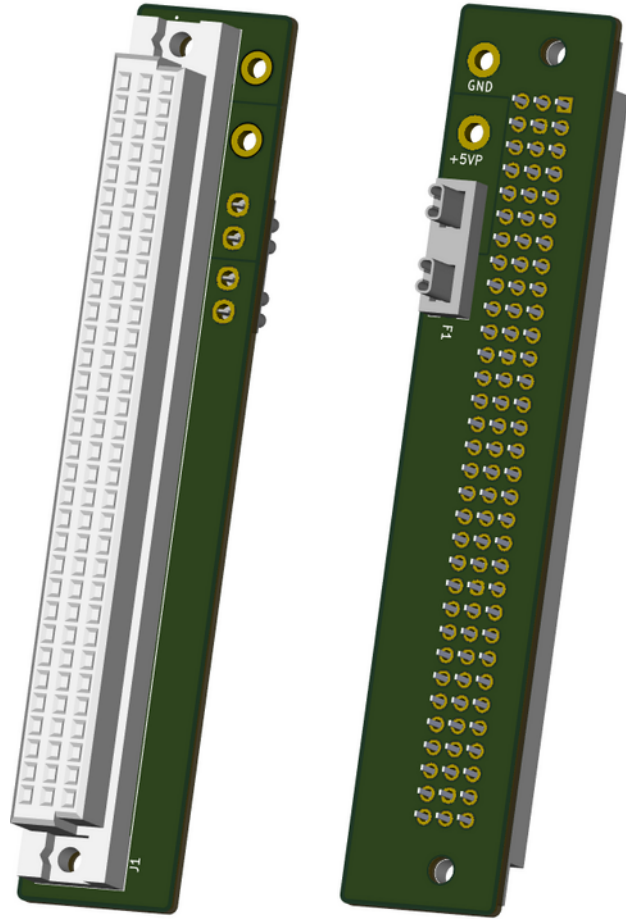
ASSEMBLING HARNESES



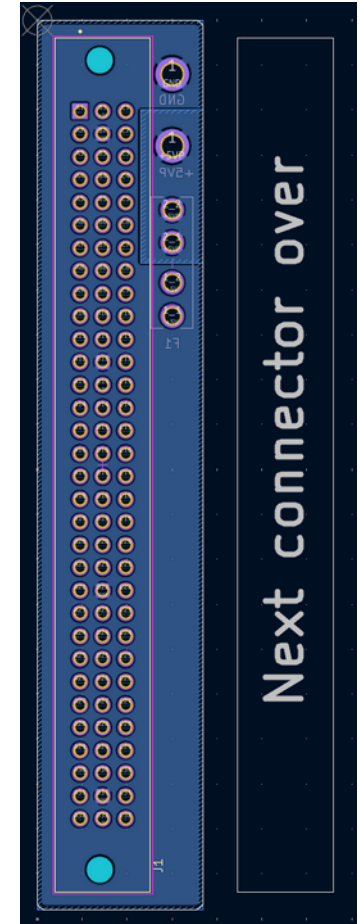
Quantities

- ☛ All four Barrel KLM crates each need two fanout boards
 - ☛ All four EBWD crates each need three fanout boards
 - ☛ All four EFWD crates each need four fanout boards
 - two fully populated (with four) and two populated with only three jumper harness assemblies for a total of 14 per crate
- ☛ Each w/bk pair of 12 AWG wire segments from the WAGO connectors to their respective fanout board can be cut about 8 cm longer than the previous pair
- ☛ This will make all of the jumper harnesses well aligned with their respective "KLM aux SRAM" boards

REAR TRANSITION MODULE (RTM) POWER SUPPLY PCB



- ☕ "+5VP" = +5 V Fuse Protected
- ☕ Plan to solder our 12 AWG w/bk wires directly to this PCB
- ☕ Layout includes layer "User.1" to show layout of next backplane connector 0.8 inches from the first →
- ☕ A 1 A fuse should suffice, but we need to double check the current draw when the SRAM is continuously being used to establish a peak value
 - 🌰 Expected draw is < 250 mA, so 1 A fuse provides an adequate safety factor



INSTALLING HARNESSES

- ☕ The 2-conductor shielded twisted 12 AWG cable needs to be cut to length prior to soldering to the RTM power supply PCB
 - ☕ I would cut the prototype to 1.2 m and check the fit
- ☕ The RTM should be soldered to the cable prior to installation
- ☕ After installation, the cable should be secured to the crate to prevent the RTM from being dislodged if some portion of the cable is snagged by someone working nearby
 - ☕ Of course snagging should be mitigated by thoughtful routing, but it's a no-brainer to properly strain relieve this cable if possible